

Unit 7 - Area

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The following B.E.S.T Standards will be covered in this unit:

MA.6.GR.2.1 Derive a formula for the area of a right triangle using a rectangle. Apply a formula to find the area of a triangle.

- *Clarification 1: Instruction focuses on the relationship between the area of a rectangle and the area of a right triangle.*
- *Clarification 2: Within this benchmark, the expectation is to know from memory a formula for the area of a triangle.*

MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.

- *Clarification 1: Problem types include finding area of composite shapes and determining missing dimensions.*
- *Clarification 2: Within this benchmark, the expectation is to know from memory a formula for the area of a rectangle and triangle.*
- *Clarification 3: Dimensions are limited to positive rational numbers.*

MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms and rhombi.

- *Clarification 1: Instruction focuses on the connection from the areas of trapezoids, parallelograms and rhombi to the areas of rectangles or triangles.*
- *Clarification 2: Within this benchmark, the expectation is not to memorize area formulas for trapezoids, parallelograms and rhombi.*

MA.7.GR.1.2 Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.

- *Clarification 1: Within this benchmark, the expectation is not to find areas of figures on the coordinate plane or to find missing dimensions.*

Unit 7, Lesson 1: Area of Triangles – Part 1



Benchmark

MA.6.GR.2.1 Derive a formula for the area of a right triangle using a rectangle. Apply a formula to find the area of a triangle.



Learning Target

- ☐ I can discover the formula for the area of a triangle.



7.1.1 Warm-Up: The Importance of Struggle

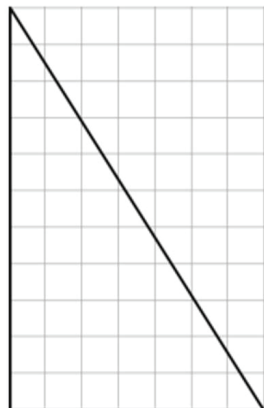
When learning something new, whether it is a sport, a musical instrument, or even math, you may experience failure before you experience success. The key is to keep going even when you make mistakes. Learning takes place during the time you practice and during the time you struggle.

1. Which choice would you feel less comfortable doing?
 - Answering a question and getting it wrong.
 - Saying, “I do not know,” as your answer to a question.
2. Describe a time that you persevered when learning something new.
3. When you persevered in learning, how did you overcome any initial fear(s)?



7.1.2 Exploration: Discovering the Area of a Right Triangle

A **right triangle** is shown. Each square of the grid paper is 1 centimeter (cm) by 1 cm.



1. Trace two copies of the right triangle on paper, then cut them out.
2. Form a rectangle using the two right triangles. Paste the triangles here.

3. Using the image just created, complete the table.

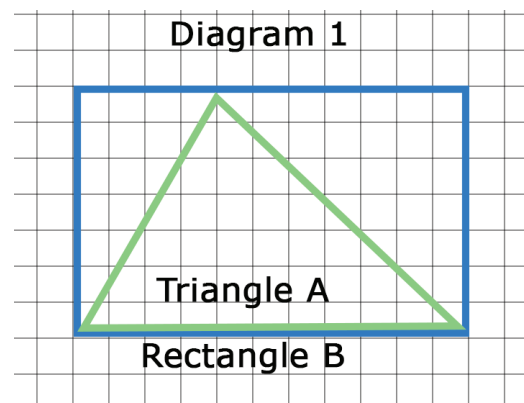
	Rectangle	Triangle
Height	_____ cm	_____ cm
Base	_____ cm	_____ cm

4. What is the area of the rectangle? Include units.
5. Describe the relationship between the area of one right triangle and the area of the rectangle.
6. What is the formula for the area of a rectangle?
7. Use the relationship observed and the formula for the area of a rectangle to write a formula that can be used to find the area of a right triangle.

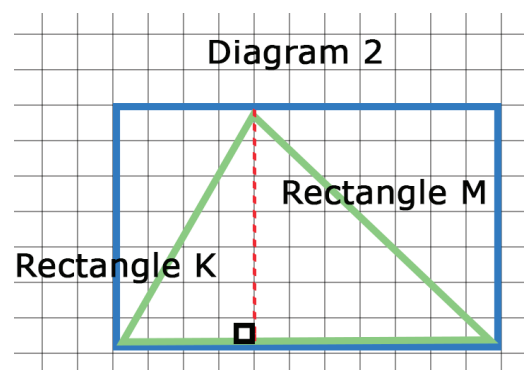


7.1.3 Collaboration: Area of an Acute and an Obtuse Triangle

An **acute triangle** is shown. Each square of the grid paper in Diagram 1 and Diagram 2 is 1 centimeter (cm) by 1 cm.



Rectangle B is drawn around acute Triangle A, as shown.



The height of acute Triangle A is shown as a dashed red line. It divides Rectangle B into two smaller rectangles, Rectangle K and Rectangle M, and divides Triangle A into two smaller right triangles.

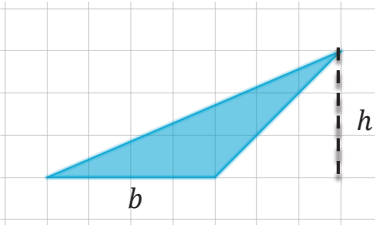
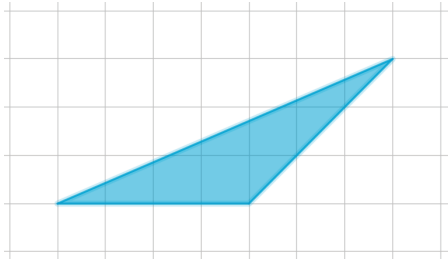
1. Use the two diagrams to complete the table.

	Area of the Rectangle	Area of the Right Triangle Drawn Inside the Rectangle
Rectangle K		
Rectangle M		

2. The area of acute Triangle A is equal to the sum of the right triangle inscribed in Rectangle K and the right triangle inscribed in Rectangle M.

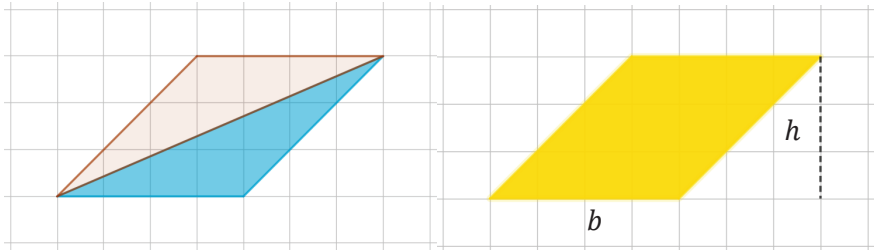
- What is the area of acute Triangle A?
- What is the area of Rectangle B?
- What is the relationship between the area of Rectangle B and the area of acute Triangle A?

3. An **obtuse triangle** is shown. Each square of the grid paper is 1 centimeter (cm) by 1 cm.

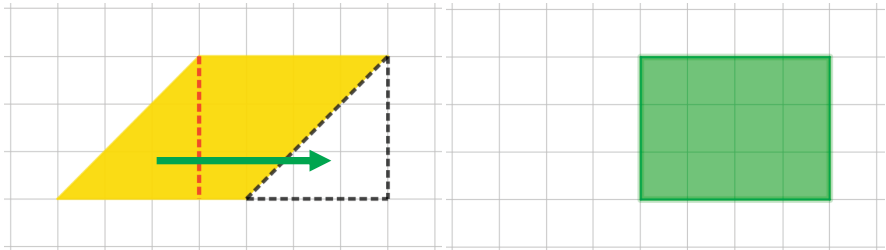


- What is the length of the triangle's base (b)?
- What is the height (h) of the triangle?

4. A copy of the triangle is made, and the two triangles are arranged to form a parallelogram with a base of 4 cm and a height of 3 cm, as shown.



The parallelogram is then decomposed, and the parts are rearranged to form a rectangle.



- Complete the table.

	Rectangle	Obtuse Triangle
Height	_____ cm	_____ cm
Base	_____ cm	_____ cm

- How many obtuse triangles were needed to create the rectangle?

c. What is the area of the rectangle?

d. What is the relationship between the area of the rectangle and the obtuse triangle?

e. Based on this relationship, what is the area of the obtuse triangle?



7.1.4 Bring It Together: Formula for the Area of a Triangle

1. Complete the statements.

The area of the **right** triangle was

☐ half
☐ the same as

 the area of the rectangle with the same base and height.

The area of the **acute** triangle was

☐ half
☐ the same as

 the area of the rectangle with the same base and height.

The area of the **obtuse** triangle was

☐ half
☐ the same as

 the area of the rectangle with the same base and height.

2. The formula for the area of any rectangle is $A = bh$.

$$A = \text{base} \cdot \text{height}$$

The formula for the area of any triangle is $A = \underline{\hspace{1cm}}bh$.

$$A = \underline{\hspace{1cm}}(\text{base} \cdot \text{height})$$





7.1.5 Wrap-Up: Reflection

1. Place an **X** on the line to indicate your mastery of today's lesson.

I can explain why the area of a triangle is half the area of a rectangle with the same base and height for right triangles, acute triangles, and obtuse triangles.

I need help.

I can't get started.

I'm getting there.

I understand and have evidence.



2. Write a brief explanation of your placement of the X.

Unit 7, Lesson 2: Area of Triangles – Part 2



Benchmark

MA.6.GR.2.1 Derive a formula for the area of a right triangle using a rectangle. Apply a formula to find the area of a triangle.

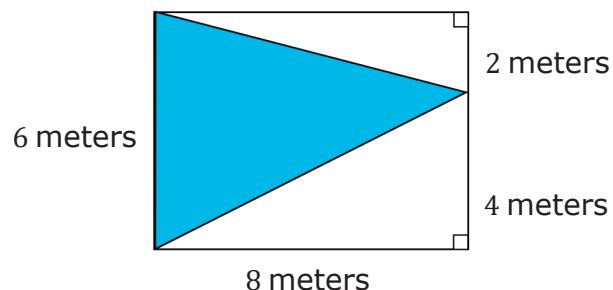


Learning Target

- ☐ I can apply a formula to find the area of triangles.



7.2.1 Warm-Up: Bell Work



1. What is the area of the rectangle? Include units.
2. What is the area of the shaded triangle? Include units.
3. Explain the relationship between the area of a rectangle and triangle that have the same height and base.



7.2.2 Guided Instruction: Find the Area of a Triangle

To find the area of any triangle, identify a base of the triangle and its corresponding height.

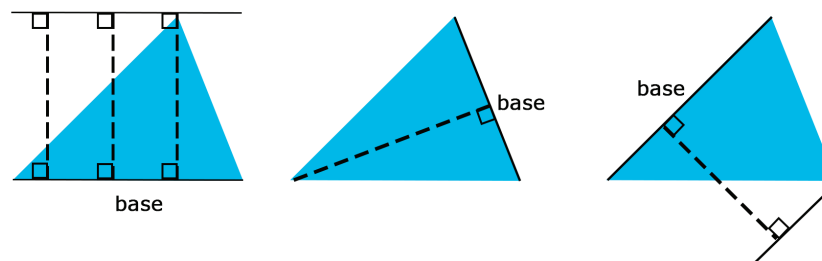


Any of the three sides of a triangle can be the **base**. The term "base" can refer to the side of a triangle and also its measurement.



The corresponding **height** of a triangle is the length of the perpendicular segment drawn from a base to its opposite vertex. The opposite vertex is the vertex of the triangle that is *not* an endpoint of the base.

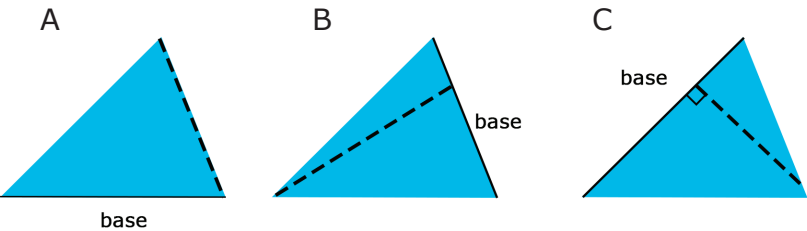
Examples of bases and their corresponding heights on copies of the same triangle are shown. Each dashed line segment represents the height of the triangle.



1. In each example the height is to the base.

- perpendicular
- opposite

2. The side used for the base is labeled on three triangles below. Complete the table to explain why each of the dashed lines do not represent the height of each triangle.



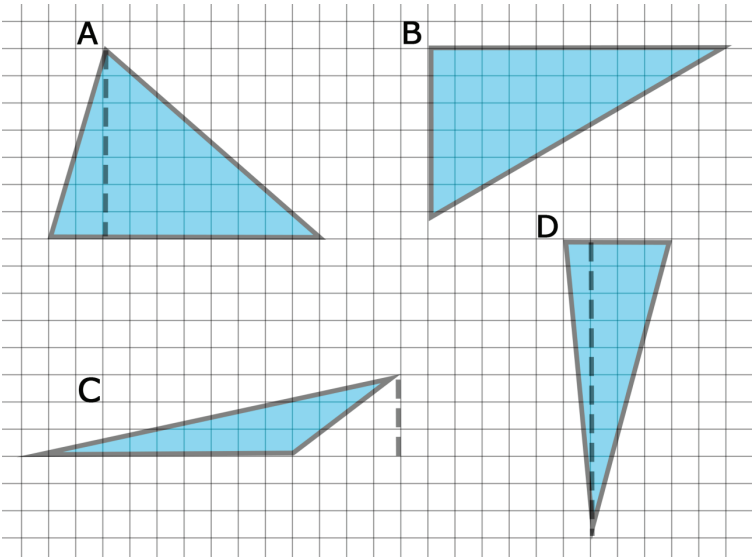
Triangle	Explanation
Triangle A	
Triangle B	
Triangle C	



Area of a Triangle $= \frac{1}{2}bh$,
where b is the base and h is the height.



3. Four triangles are shown on the grid. The side length of each square on the grid is 1 centimeter (cm).



a. For each triangle, label the line segment that is drawn to represent the height, h . Then, label the side that is the base, b .

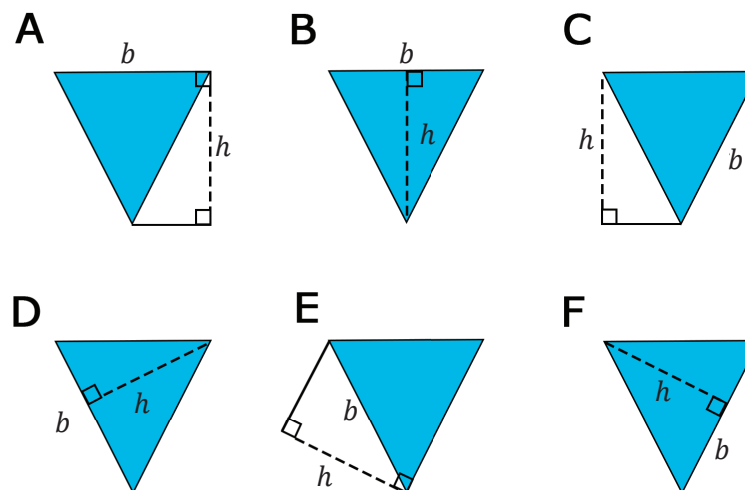
- b. In the table, record the measurement for the base and corresponding height for each triangle. Then, find the area of each triangle. Each square on the grid represents 1 centimeter (cm) by 1 cm.

Triangle	Base (cm)	Height (cm)	Area (cm ²)
Any Triangle	b	h	$\frac{1}{2}bh$
Triangle A			
Triangle B			
Triangle C			
Triangle D			



7.2.3 Guided Practice: Find the Area of a Triangle

1. Discuss with your partner which drawings correctly label base, b , and its corresponding height, h . Circle all the correct drawings.



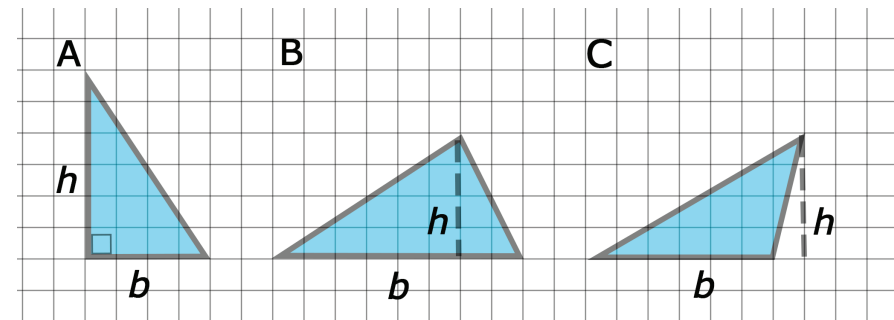
2. For each triangle below, circle the base measurement and put a square around the height measurement. Then, use the base and height to find the area of each triangle. Show your work. Include units.

	Area = _____
	Area = _____
	Area = _____
	Area = _____



7.2.4 Your Turn!

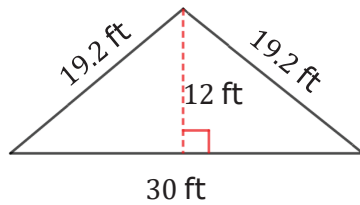
1. For each triangle, a base and its corresponding height are labeled. Complete the table by finding the area of each triangle. The side length of each square on the grid is 1 inch (in.).



Triangle	Area (in ²)
A	
B	
C	

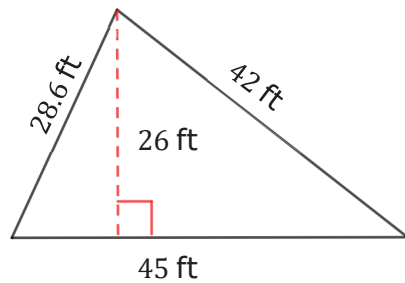
2. Find the area of each triangle. Include units.

a.



Area = _____

b.

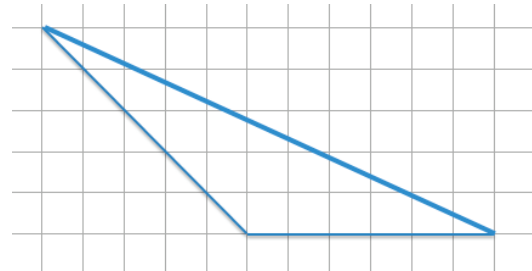


Area = _____



7.2.5 Wrap-Up: Ticket Out the Door

1. A diagram of a triangle is shown on a grid. The side length of each square on the grid represents 1 foot (ft).



- a. Label a base and its corresponding height for the triangle.
- b. Find the area of the triangle. Include units.

Unit 7, Lesson 3: Decomposing Quadrilaterals to Find Area



Benchmark

MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.



Learning Target

- ☐ I can find the area of quadrilaterals by decomposing them into triangles and rectangles.



7.3.1 Warm-Up: Perimeter and Area of a Rectangle

1. A rectangle is shown with side lengths given in centimeters (cm).

2.8 cm



7.2 cm

- a. Find the perimeter of the rectangle. Include units.

Perimeter = _____

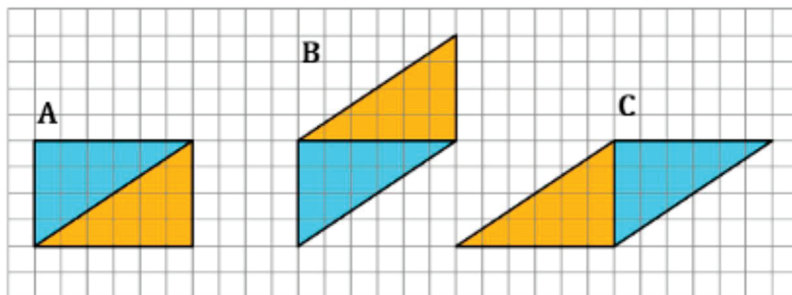
- b. Find the area of the rectangle. Include units.

Area = _____



7.3.2 Collaboration: Decomposing Parallelograms

Three parallelograms are shown on the grid. Each square is 1 centimeter (cm) by 1 cm.



1. Work with your partner to complete the table using the figures shown in the grid. Include units.

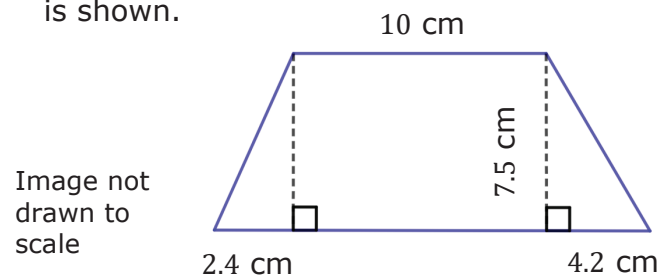
Parallelogram	Area of Blue Triangle	Area of Gold Triangle	Sum of the Two Triangles
A			
B			
C			

2. Discuss with your partner how the areas of the triangles are related to the area of the parallelogram they form. Write a brief summary of your discussion.



7.3.3 Exploration: Finding Area by Decomposing

1. A quadrilateral with measurements in centimeters (cm) is shown.

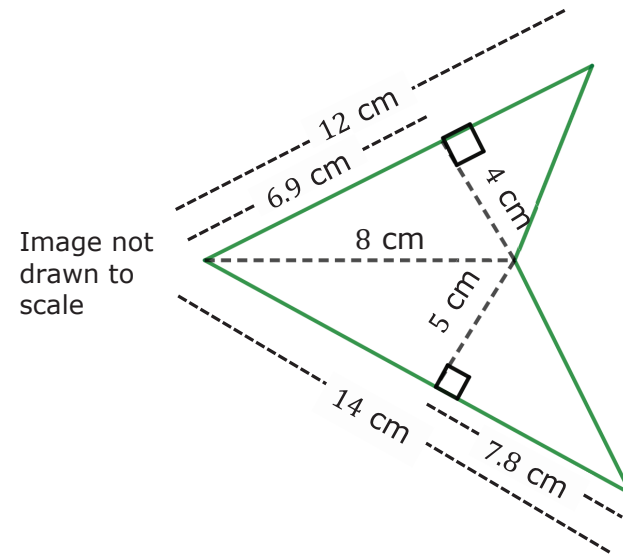


- a. Copy the image on a piece of paper.
- b. Decompose the quadrilateral into two triangles and a rectangle by cutting along the dotted lines. Glue or tape the three figures here.

- c. Using the measurements given in the original image, label the base and height of each figure.
- d. What is the area of each figure? Include units.

- e. Find the area of the original quadrilateral by finding the sum of the three figures. Include units.

2. Another quadrilateral is shown.



- a. Decompose the quadrilateral into triangles. Sketch the decomposed triangles here.
- b. Using the measurements given in the original image, label the base and height of each triangle.

- c. Find the area of the original quadrilateral by finding the sum of the areas of the decomposed triangles. Include units.

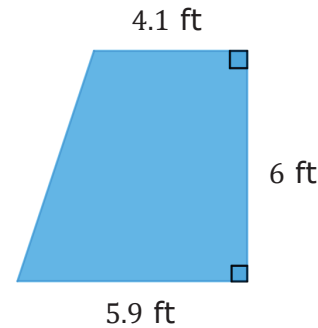
- d. Sketch any other ways you could have decomposed the original figure.

- e. Explain if decomposing the figure changes the area of the original figure.



7.3.4 Guided Practice: Finding Area by Decomposing

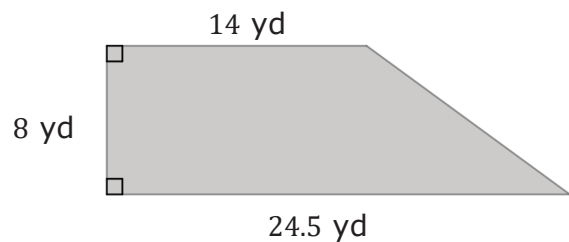
1. Decompose the quadrilateral to find the area. Lengths are shown in feet (ft). Include units.



Area = _____

Image not drawn to scale

2. The floor plan of a store is shown with measurements in yards (yd).

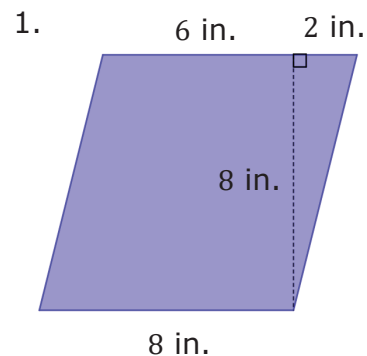


- a. The store found 160 square yards (yd^2) of tile on sale. Based on the floor plan drawn, will the store have enough tiles to redo the flooring if they buy what is on sale?
- b. Determine the amount of tile, in yd^2 , the store is short or has left over.



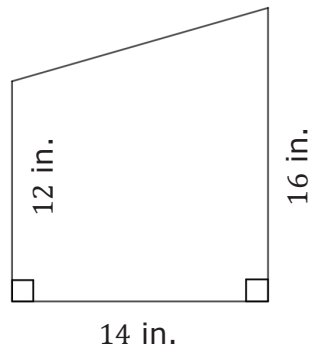
7.3.5 Your Turn!

Find the area of each quadrilateral by decomposing. Lengths are shown in inches (in.). Include units.



Area = _____

2.



Area = _____

3.

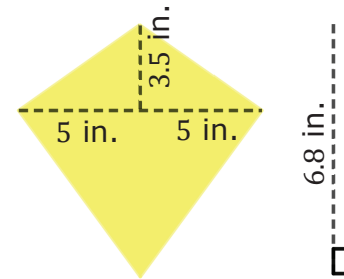


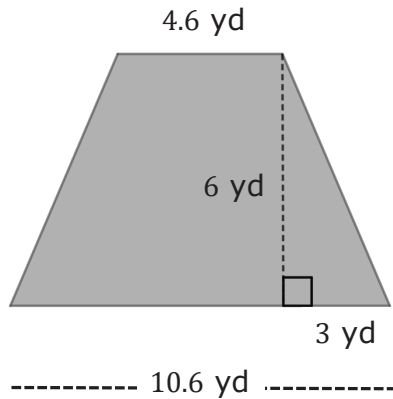
Image not drawn to scale

Area = _____



7.3.6 Wrap-Up: Ticket Out the Door

1. Find the area of the quadrilateral by decomposing. Lengths are given in yards (yd). Include units.



Area = _____

Unit 7, Lesson 4: Deriving Area Formulas for Trapezoids, Parallelograms, and Rhombi



Benchmark

MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.



Learning Target

- ☐ I can use the formulas for the area of triangles and rectangles to derive formulas for the area of trapezoids, parallelograms, and rhombi.



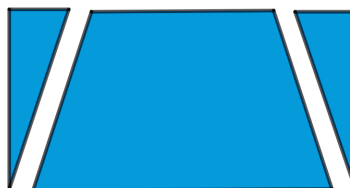
7.4.1 Warm-Up: Notice and Wonder

Two figures are shown.

Figure 1



Figure 2



1. What do you notice about these two figures?

2. What do you wonder about the figures?

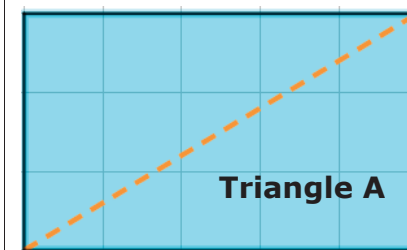


7.4.2 Refresher: Decomposing to Find the Area of a Triangle

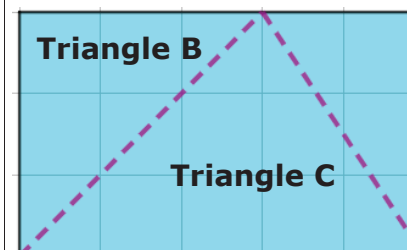
Figures can be decomposed into smaller pieces that are then rearranged and composed into other shapes. Decomposing is useful when finding the area of a figure.

1. Complete the table.

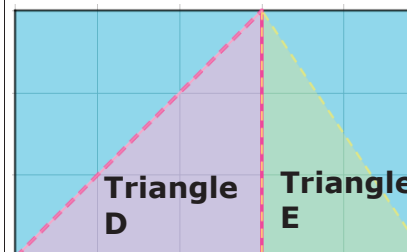
The rectangle is divided into _____ triangles. The area of Triangle A is _____ of the area of the rectangle.



The rectangle is divided into _____ triangles. The area of Triangle B is _____ of the area of the rectangle. The area of Triangle C is _____ of the area of the rectangle.



The rectangle is divided into _____ triangles. The area of Triangle D is _____ of the area of the rectangle. The area of Triangle E is _____ of the area of the rectangle.





7.4.3 Activity: Deriving the Formula for the Area of a Parallelogram



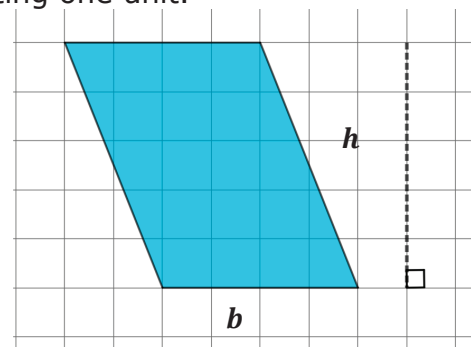
A **quadrilateral** is a polygon with exactly four sides and four vertices.

A parallelogram is a type of quadrilateral. Not only does a parallelogram have four sides and four vertices, but its opposite sides are also parallel.



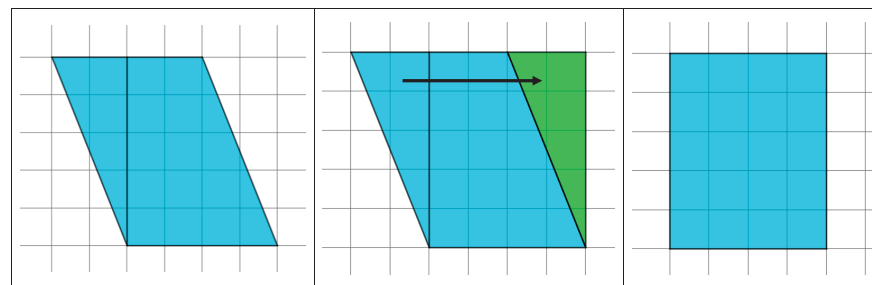
A **parallelogram** is a quadrilateral containing two pairs of parallel sides.

1. A parallelogram is shown on a grid with each gridline representing one unit.



Complete each statement.

- a. The base of the parallelogram is _____ units.
 - b. The height of the parallelogram is _____ units.
2. The images below show the same parallelogram being decomposed and then rearranged to form a rectangle.



Complete each statement.

- a. The base of the rectangle is _____ units.
- b. The height of the rectangle is _____ units.

3. Discuss with your partner the relationship between the area of the parallelogram and the area of the rectangle. Write a summary of your discussion.

4. Complete the statements.

A parallelogram can be decomposed and rearranged into a

☐ triangle
☐ rectangle with the same base and height. The

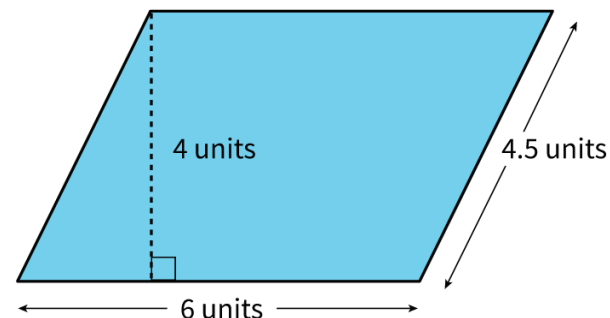
formula used to find the area of a rectangle is

- ☐ $A = bh.$
☐ $A = \frac{1}{2}bh.$

Therefore, the area of the parallelogram is

- ☐ $A = bh.$
☐ $A = \frac{1}{2}bh.$

5. A parallelogram with its dimensions labeled is shown.



- a. Fill in the blanks to find the area of the parallelogram.

Area of parallelogram = $b \cdot h$

Area = _____ $\cdot$ _____

Area = _____ units²

- b. Three lengths are shown in the figure. With your partner, discuss the length that was not used to find the area of the parallelogram. Then, write an explanation for why this length was not used.



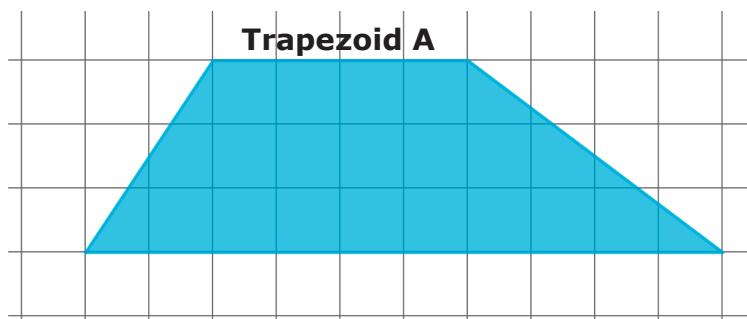
7.4.4 Activity: Deriving the Formula for the Area of a Trapezoid

A trapezoid is another type of quadrilateral.



A **trapezoid** is a quadrilateral with at least one pair of parallel sides.

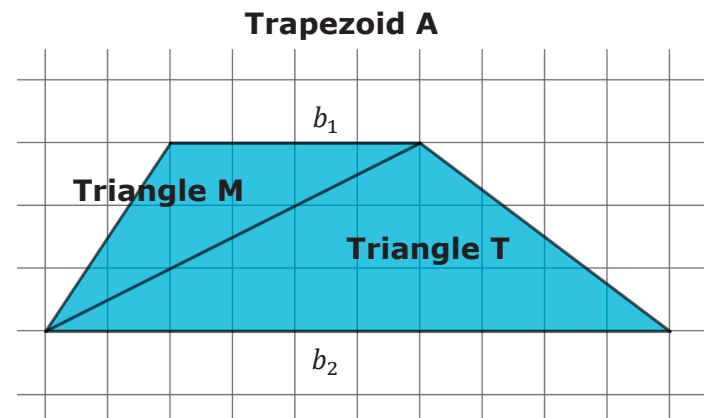
Trapezoid A is shown on a grid with each gridline representing one unit.



The two parallel sides of a trapezoid are both called bases. To distinguish between the two bases, one is named base one, b_1 , and the other base two, b_2 .

1. On Trapezoid A, label one of the bases b_1 and the other b_2 .
2. Complete each statement.
 - a. The length of b_1 of Trapezoid A is _____ units.
 - b. The length of b_2 of Trapezoid A is _____ units.
 - c. The height of Trapezoid A is _____ units.

3. Trapezoid A is decomposed into two triangles and labeled, as shown below.



- a. Complete the statements.

The base of Triangle M, b_1 , is _____ units, and its corresponding height is _____ units.

The base of Triangle T, b_2 , is _____ units, and its corresponding height is _____ units. When a trapezoid is decomposed into two triangles by drawing a diagonal, the height of both triangles will always be the same.

- b. Davita wrote a formula for the area of Trapezoid A.

Davita's Equation

Area Trapezoid A = Area Triangle M + Area Triangle T

$$A = \frac{1}{2}b_1h + \frac{1}{2}b_2h$$

Explain how Davita determined this formula.

- c. The formula Davita wrote can be used to find the area of any trapezoid. Complete the steps described in the table to rewrite the formula.

Area of Trapezoid A	
Davita's formula	$A = \frac{1}{2}b_1h + \frac{1}{2}b_2h$
Use the distributive property to rewrite with a common factor h	$A = \frac{1}{2}h(b_1 + b_2)$
Use the distributive property to rewrite with a common factor $\frac{1}{2}$	$A = \frac{1}{2}h(b_1 + b_2)$



7.4.5 Activity: Deriving the Formula for the Area of a Rhombus

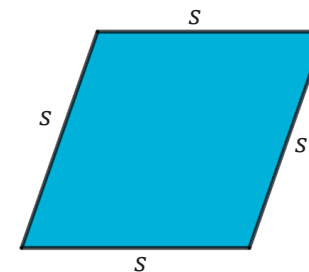
A rhombus is another type of quadrilateral.



A **rhombus** is a quadrilateral containing four sides of equal length.

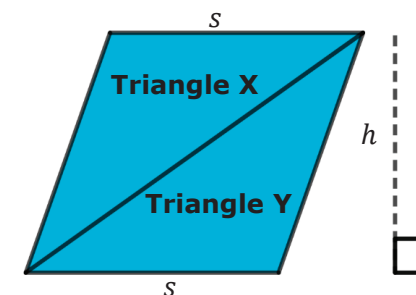
Rhombus F is shown with four congruent sides.

Rhombus F



Rhombus F is decomposed into two triangles, Triangle X and Triangle Y, as shown.

Rhombus F



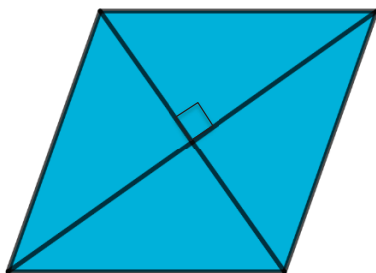
1. Felicia wrote expressions to represent the areas of the two triangles.

Area of Triangle X	Area of Triangle Y
$\frac{1}{2} \cdot s \cdot h$	$\frac{1}{2} \cdot s \cdot h$

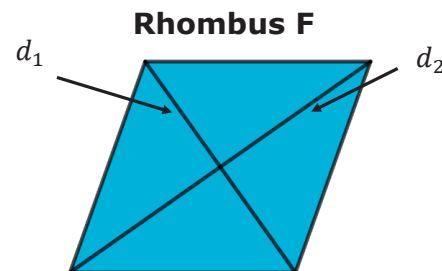
Discuss with your partner whether you agree with the expressions Felicia wrote. Write a summary of your discussion.

The diagonals of a rhombus can also be used to find its area.

The two diagonals of a rhombus are perpendicular to each other and divide each other in half.



2. The two diagonals, d_1 and d_2 , of Rhombus F are drawn and labeled below.



Mark on the figure the two lengths that are equal to $\frac{1}{2}d_1$ and the two lengths that are equal to $\frac{1}{2}d_2$.

3. The first column of the table shows Rhombus F with parts highlighted. Use the lengths marked on Rhombus F in the previous question to fill in the blanks to find the area of the triangles.

Triangle	Area
Rhombus F 	$\text{Area} = \frac{1}{2}(b)(h)$ $\text{Area} = \frac{1}{2}(\text{ })(\text{ })$ Rewrite the area by multiplying any fractions. $\text{Area} = \text{ $
Rhombus F 	$\text{Area} = \frac{1}{2}(b)(h)$ $\text{Area} = \frac{1}{2}(\text{ })(\text{ })$ Rewrite the area by multiplying any fractions. $\text{Area} = \text{ $

4. Compare your work with your partner and resolve any differences, if necessary.

5. Work with your partner to express the area of Rhombus F as the sum of the area of the yellow triangle and the pink triangle found in the table.

Area of Rhombus F = _____

6. Use words, values, or expressions from the bank shown to complete the statements. Items from the bank may be used more than once.

$\frac{1}{2}$

sum

$\frac{1}{4}(d_1)(d_2)$

diagonal

Drawing a _____ of a rhombus divides the rhombus into two identical triangles. The area of the rhombus is equal to the _____ of the areas of these two triangles. If one of these triangles is equal to _____, then two of these triangles equal $2 \cdot$ _____. Therefore, the formula for the area of a rhombus using its diagonals can be written as $A = \underline{\hspace{1cm}}(d_1)(d_2)$.



7.4.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point for you in today's lesson on deriving the formula of parallelograms, trapezoids, and rhombi?



Unit 7, Lesson 5: Finding the Area of Trapezoids, Parallelograms, and Rhombi



Benchmark

MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.



Learning Target

- ☐ I can find the area of trapezoids, parallelograms, and rhombi.



7.5.1 Warm-Up: A Day in the Life

Betsy McIver, a mechanical engineer, works in the Walt Disney Imagineering area and designs systems for the interactive attractions at the theme park. She mentions that her job can include creative left-brain thinking, as well as technical right-brain thinking.



She says that to be successful at her job, it is important to understand math, design, and theatre. She also mentions that many disciplines, such as designers and architects, all work together to make everything possible in the entertainment engineering industry.

1. Describe a ride you saw that looked exciting and entertaining.



7.5.2 Guided Practice: Area of Parallelograms, Trapezoids, and Rhombi

- Complete the table to match each type of quadrilateral with the formula to find its area.

$$A = \frac{1}{2}h(b_1 + b_2)$$

$$A = \frac{1}{2}d_1d_2$$

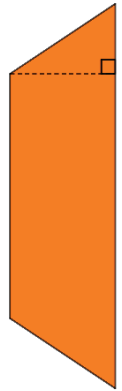
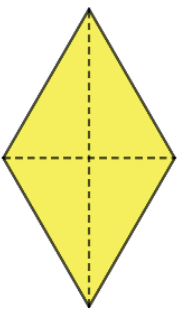
$$A = bh$$

Quadrilateral	Formula for the Area
parallelogram	
trapezoid	
rhombus	

- The dimensions of three quadrilaterals are given in the table. Label each figure shown with the given dimensions in centimeters (cm). Then, calculate the area of each figure using the appropriate formula. Show your work.

Figure	Dimensions	Area
Parallelogram 	$b = 22 \text{ cm}$ $h = 8 \text{ cm}$	

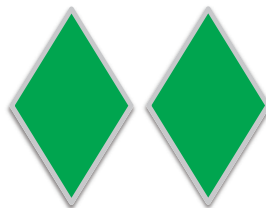


Figure	Dimensions	Area
Trapezoid 	$b_1 = \frac{7}{12} \text{ cm}$ $b_2 = \frac{11}{12} \text{ cm}$ $h = \frac{1}{4} \text{ cm}$	
Rhombus 	$d_1 = 6.5 \text{ cm}$ $d_2 = 12.25 \text{ cm}$	

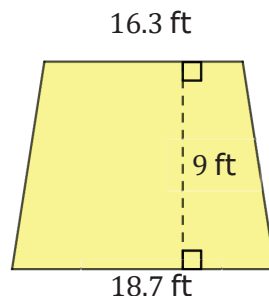


7.5.3 Your Turn!

1. Alina is making rhombus-shaped earrings with green stones. The measure of one diagonal of each earring is $1\frac{3}{8}$ centimeters (cm), and the length of the other diagonal measures $1\frac{4}{9}$ cm. What is the area, in square centimeters (cm^2), of two earrings?

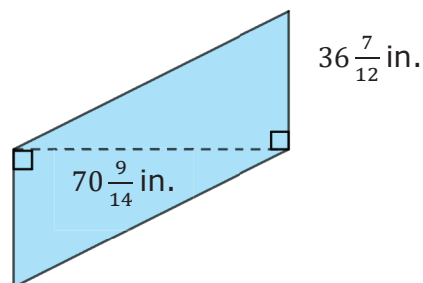


2. One wall of Kel's apartment is shaped like a trapezoid with one base 18.7 feet (ft), the other base 16.3 ft, and a height of 9 ft.

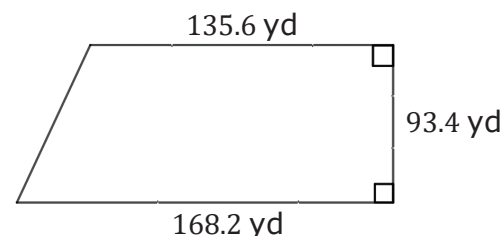


Kel plans to paint this wall blue and needs to know how much paint to buy. What is the area of the wall Kel plans to paint?

3. Mika, a designer, is creating a geometric design in a tile floor. The design will be in the shape of a parallelogram with $36\frac{7}{12}$ inch (in.) bases and a height of $70\frac{9}{14}$ in. How many square inches of tile will she need for the design?



4. The Smart family's horse pasture is in the shape of a trapezoid. One of the parallel sides is 168.2 yards (yd) long and the other is 135.6 yd long. The distance between these sides is 93.4 yd.



What is the area of the horse pasture?

5. The Dockland Building in Hamburg, Germany is shaped like a parallelogram.

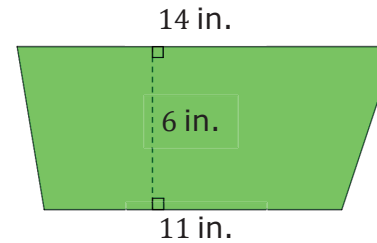


If the base of the building is 86 meters (m) and its height is 47 m, what is the area of the face of the building shown in the image?



7.5.4 Wrap-Up: Ticket Out the Door

1. Don's work for finding the area of the trapezoid is shown below.



Don's Work

$$A = \frac{1}{2} h(b_1 + b_2)$$

$$A = \frac{1}{2}(14)(11 + 6)$$

$$A = \frac{1}{2}(14)(17)$$

$$A = \frac{1}{2} \cdot 238$$

$$A = 119 \text{ in}^2$$

- a. Write Don a note explaining what he did correctly and where he made an error.
- b. What is the area of the trapezoid?

Unit 7, Lesson 6: Area of Composite Figures



Benchmark

MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.



Learning Targets

- ☐ I can find the area of composite figures.
- ☐ I can find the value of a missing dimension for a quadrilateral or a composite figure.



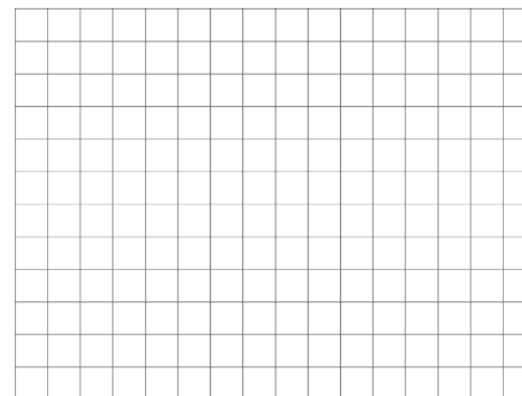
7.6.1 Warm-Up: Closest to Six

- Use the digits 0 to 9, at most one time each, to fill in ordered pairs for all three points, such that the area of Triangle ABC is closest to 6 square units.

A (,)

B (,)

C (,)



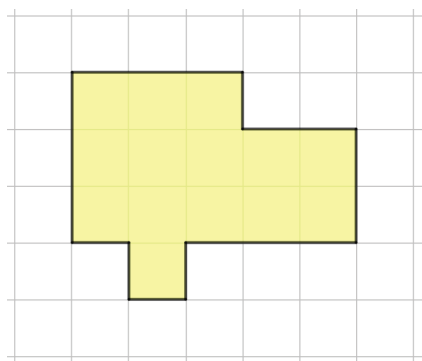


7.6.2 Guided Instruction: Area of Composite Figures



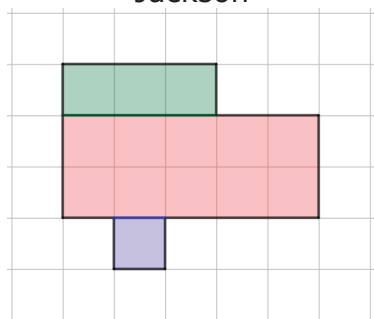
A **composite figure** is a two- or three-dimensional figure that can be decomposed into smaller figures.

Jackson and Danna were instructed to find the area of the composite figure below. The length of each square is 1 centimeter (cm).

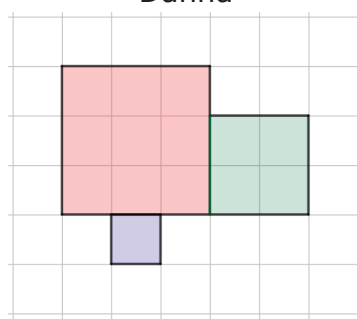


Both Jackson and Danna decomposed the figure, but their teacher noticed they each decomposed it differently.

Jackson



Danna



1. Complete the table about each student's work. Include units.

Student	Area of Red Rectangle	Area of Blue Rectangle	Area of Green Rectangle
Jackson			
Danna			

2. To find the area of the composite figure, find the sum of the areas of the rectangles for each student. Include units.

	Area of the Composite Figure
Jackson	
Danna	

3. Explain why both students got the same area of the composite shape.

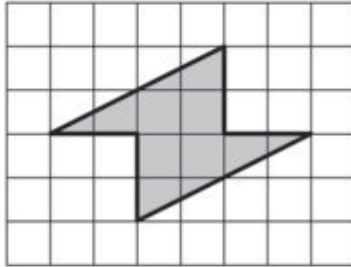


The **area of a composite figure** is the sum of the areas of the figure's decomposed parts.



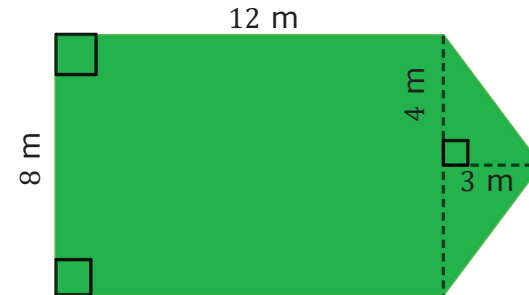
7.6.3 Guided Practice: Area of Composite Figures

1. A composite shape is shown. The length of each square is 1 inch (in).



- Draw lines to decompose the figure into rectangles and/or triangles.
- Find the area of each rectangle and/or triangle. Include units.
- Find the sum of the areas of the rectangles and/or triangles to find the area of the composite figure. Include units.
- Find a classmate who decomposed the shape differently than you did. Describe their process.

2. Thomas has decided to put grass sod in his backyard. A drawing of his backyard is shown.



How much sod does Thomas need to buy?

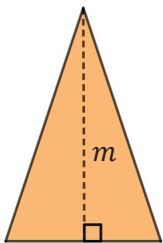
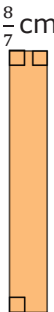


7.6.4 Collaboration: Finding a Missing Value

- Complete the table by filling in the blanks for the following math facts.

If...	Then...
$40 \div 8 = \underline{\hspace{2cm}}$	$8 \times \underline{\hspace{2cm}} = 40$
$27 \div 3 \div \frac{1}{2} = \underline{\hspace{2cm}}$	$\frac{1}{2} \times \underline{\hspace{2cm}} \times 3 = 27$
$8 \div \frac{8}{7} = \underline{\hspace{2cm}}$	$\underline{\hspace{2cm}} \times \frac{8}{7} = 8$

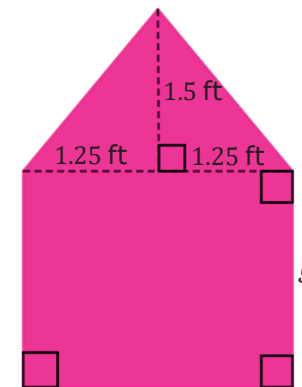
- With your partner, use the previous math facts to find the missing dimensions of the quadrilaterals. All measurements are in centimeters (cm).

		
Area = 40 cm^2	Area = 27 cm^2	Area = 8 cm^2
$k = \underline{\hspace{2cm}}$	$m = \underline{\hspace{2cm}}$	$x = \underline{\hspace{2cm}}$



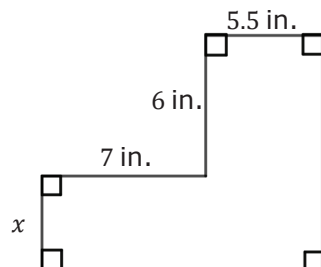
7.6.5 Guided Practice: Finding the Value of the Missing Dimension

- Xavier is painting the back wall of a dollhouse. He knows the back wall of the dollhouse has an area of 6.875 square feet (ft^2).



- What is the area of the triangular top? Include units.
- What is the area of the rectangular base? Include units.
- What is the height, g , of the rectangular base? Include units.

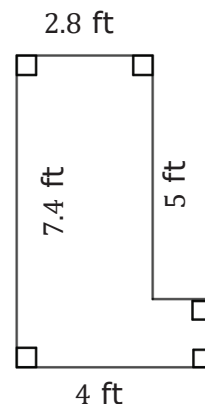
2. The area of the composite figure is 84 square inches (in^2). Find the missing dimension. Include units.



7.6.6 Your Turn!

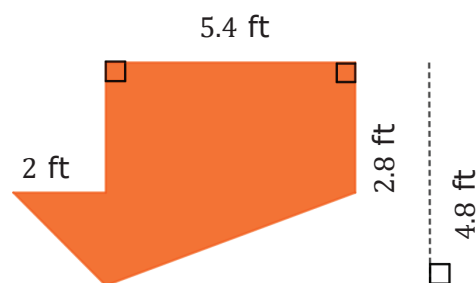
For each composite figure, find the area, in square feet (ft^2). Show your work.

1.



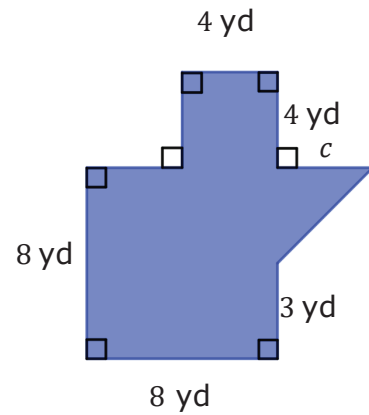
Area = _____

2.



Area = _____

3. The area of the composite figure is 87.5 square yards (yd^2). Find the value of the missing dimension. Include units.



Not drawn to scale



7.6.7 Wrap-Up: Reflection

Complete each statement based about today's lesson.

1. I am still unsure about ...

2. One question I still have about today's lesson is ...

Unit 7, Lesson 7: Finding the Area of Polygons



Benchmark

MA.7.GR.1.2 Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.



Learning Targets

- ☐ I can decompose a regular polygon into triangles to find the area.
- ☐ I can decompose a regular polygon into triangles and quadrilaterals to find the area.



7.7.1 Warm-Up: Which One?

The dimensions of two rectangles are given.

Rectangle A

Rectangle B

13 units by 15 units

2 units by 3 units

1. Explain which rectangle is more like a square.

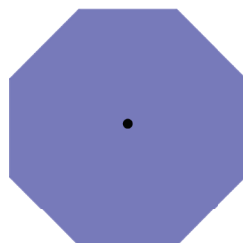


7.7.2 Guided Instruction: Decomposing Regular Polygons



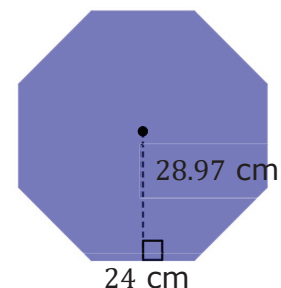
A **regular polygon** is a polygon that is both equilateral (all sides congruent) and equiangular (all angles congruent).

1. A regular octagon is shown.



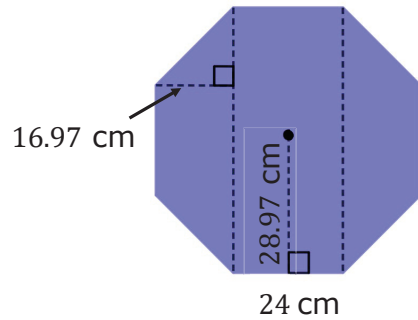
- On the figure, mark all congruent angles and all congruent sides.
- Draw lines to decompose the regular octagon into congruent triangles.
- How many congruent triangles did you decompose the regular octagon into?
- Discuss with your partner how decomposing the octagon into congruent triangles is helpful in determining the area.

2. Some dimensions of an octagon are given in centimeters (cm). Each side of the octagon has a length of 24 cm. The perpendicular distance from the center to the midpoint of a side is 28.97 cm.



- Find the area of one decomposed triangle.
- Find the area of the regular octagon to the nearest hundredth of a square centimeter (cm^2) by finding the sum of the areas of all the decomposed triangles.

3. An octagon is divided into two trapezoids and a rectangle. Each side of the octagon has a length of 24 cm. The perpendicular distance from the center to the midpoint of a side is 28.97 cm. The height of each trapezoid is 16.97 cm.



- a. Find the area of each decomposed quadrilateral to the nearest hundredth.
- b. Using the decomposed shapes, determine the area of the octagon to the nearest hundredth of a square centimeter (cm^2).

4. Complete the chart using the areas from 2b and 3b

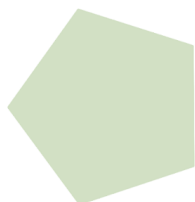
Decomposition	Area of Octagon
triangles	
quadrilaterals	

5. What do you notice about the areas?



7.7.3 Collaboration: Decomposing Regular Polygons to Find Area

- Decompose each of the following regular polygons.



- Share with your partner how you decomposed your shapes. If your partner decomposed the shapes differently, draw on the blue shapes how they decomposed their shapes. If you decomposed the shapes in the same way, work together to create a different way to decompose the shapes.



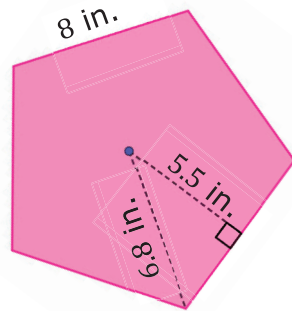
- The table shows a pentagon and a hexagon. Given measurements in centimeters (cm). Work with your partner to determine the area rounded to the nearest hundredth.

Pentagon	Hexagon
Side lengths = 20 cm	Side lengths = $9\frac{1}{4}$ cm
Area of Pentagon	Area of Hexagon



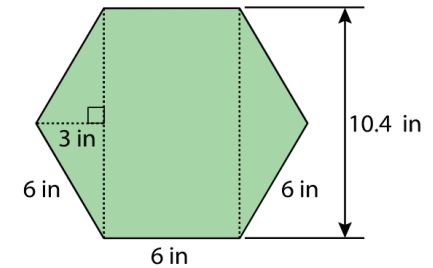
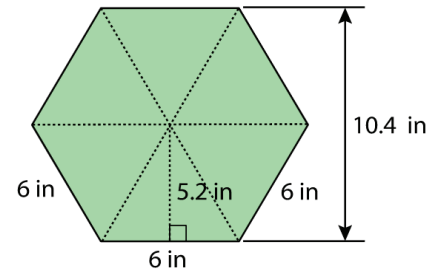
7.7.4 Your Turn!

1. A regular pentagon is shown with lengths in inches (in.).



What is the area of the pentagon?

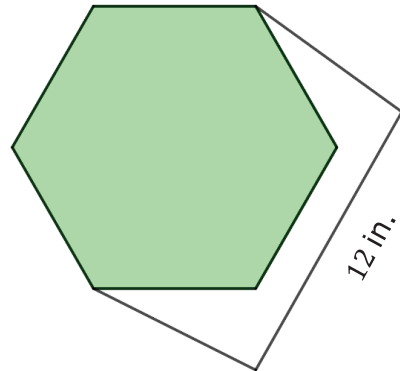
2. Lin and Andre used different methods to find the area of a regular hexagon with 6 in. sides. Lin decomposed the hexagon into six identical, equilateral triangles. Andre decomposed the hexagon into a rectangle and two triangles.



- a. Explain why Lin and Andre will find the same area of the hexagon.
- b. Find the area of the hexagon using both Lin's and Andre's methods.

Lin's Method	Andre's Method

- c. Decompose the shape differently than Lin and Andre did.

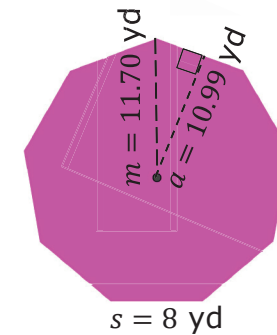


- d. Using the measurements given in all three figures, find the area by finding the sum of your decomposed parts.



7.7.5 Wrap-Up: Error Analysis

1. Vernon found the area of the nonagon. His work is shown.



Vernon's Work

$$\begin{aligned} \text{Area of Triangle} \\ A &= \frac{1}{2}(8)(11.70) \\ A &= 46.8 \text{ yd}^2 \end{aligned}$$

$$\begin{aligned} \text{Area of Nonagon} \\ A &= 18(46.8) \\ A &= 842.4 \text{ yd}^2 \end{aligned}$$

- a. Explain to Vernon what error(s) he made in his work.
- b. What is the area of the nonagon?

Unit 7, Lesson 8: Finding the Area of Composite Figures



Benchmark

MA.7.GR.1.2 Solve mathematical or real-world problems involving the areas of polygons or composite figures by decomposing them into triangles or quadrilaterals.



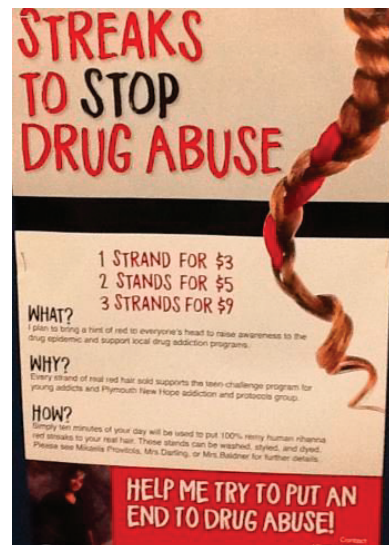
Learning Target

- ☐ I can decompose composite figures into triangles and quadrilaterals to find the area.



7.8.1 Warm-Up: Math Fail

The advertisement below is for the sale of colored strands for braids to raise money for a Stop Drug Abuse campaign.



1. Explain what the math fail is in the advertisement.

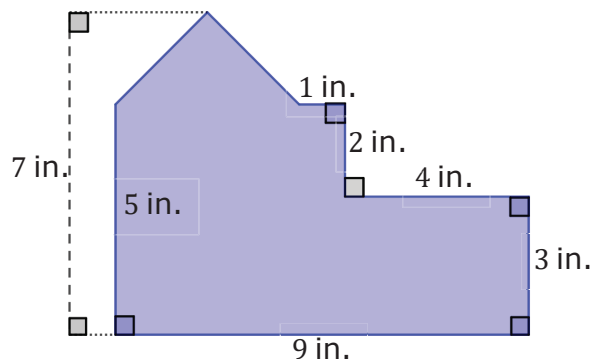


7.8.2 Guided Instruction: Finding the Area of Polygons



Composite figures are two- or three-dimensional figures that can be decomposed into smaller figures.

A floorplan is shown. All segment lengths are in inches (in.).



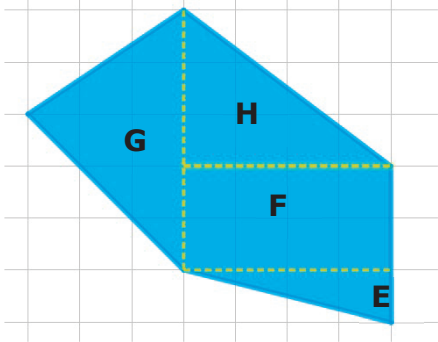
1. With your partner, decompose the floorplan into triangles and quadrilaterals.
2. Find the area of the floorplan. Show your work.

3. A polygon inscribed inside a rectangle is shown on the grid. Each grid line is 1 unit.

a. Complete the table.

Polygon Inscribed in a Rectangle	
Decomposition of Rectangle	
Area of the large rectangle	
Area of R	
Area of S	
Area of T	
Area of W	
Area of Y	
Area of the blue polygon	

- b. The same polygon is shown on the grid. Each grid line is 1 unit.

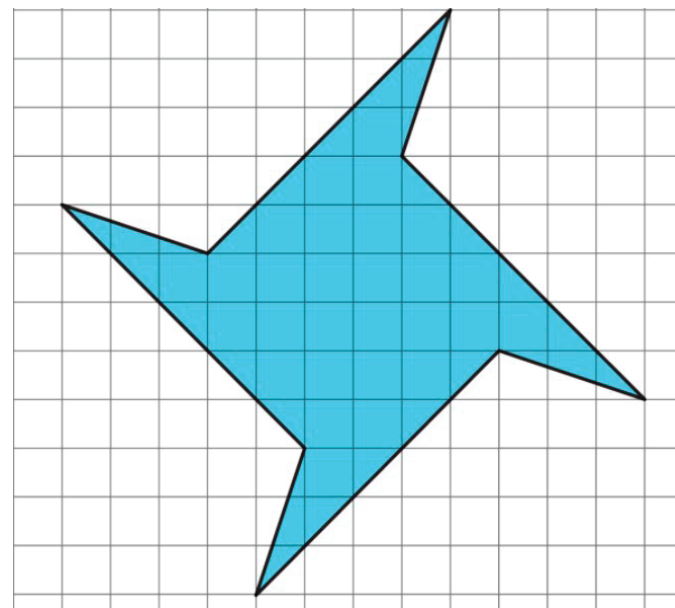
Polygon

Decomposition of Polygon
Area of E
Area of F
Area of G
Area of H
Area of the blue polygon

- c. Discuss with your partner the different possible methods of finding the area of the blue polygon. List one strength of each method.



7.8.3 Collaboration: Area of a Polygon

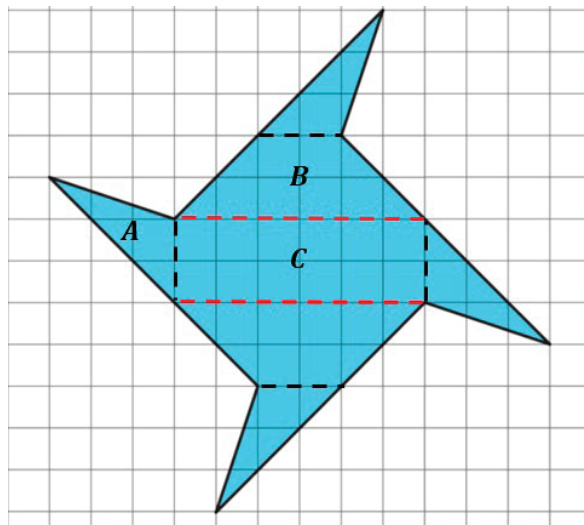
1. A polygon is shown on grid paper. Each grid line is 1 unit.



- Decompose the polygon into triangles and quadrilaterals.
- Share your decomposition with your partner. List at least one similarity and one difference.

2. Taahira decomposed the polygon into 7 pieces using 3 different shapes, as shown. Each grid line is 1 unit.

Work with your partner to find the area of each shape and the total area of the polygon.

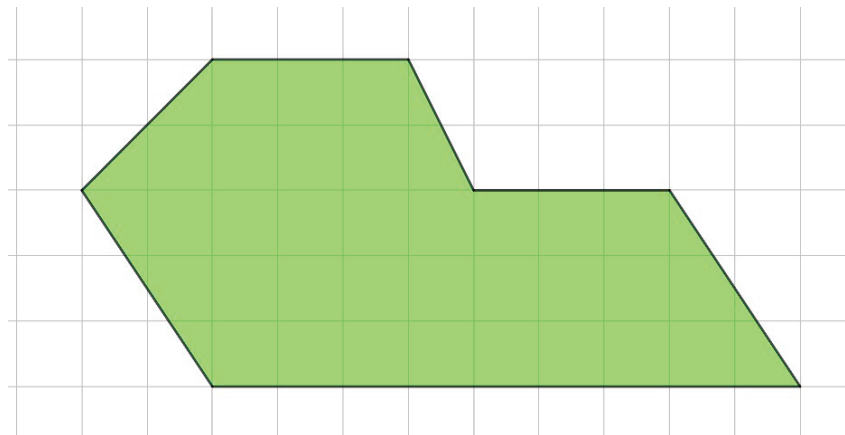


Shape	Type of Shape	Area of the shape	Number of Shapes	Total Area
A			4	
B				
C			1	
Area of the Polygon				



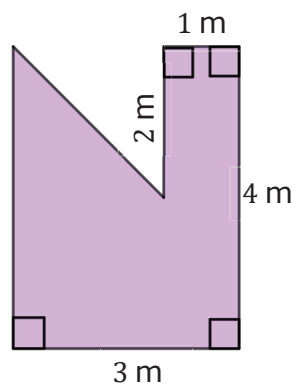
7.8.4 Guided Practice: Decomposing Composite Figures to Find Area

1. A polygon is shown on a grid. Each grid line is 1 unit.



Find the area of the polygon.

2. A polygon is shown with the given measurements in meters (m).

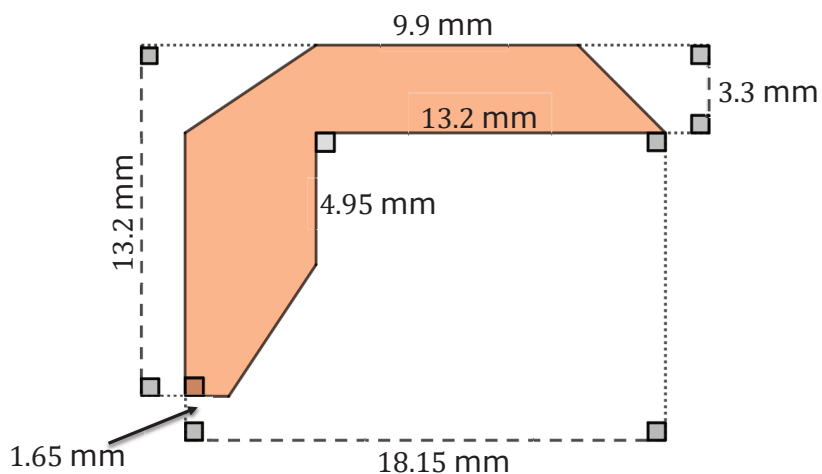


What is the area of the composite figure?



7.8.5 Your Turn!

1. Find the area of the polygon. Lengths are shown in millimeters (mm).





7.8.6 Wrap-Up: Reflection

1. Place an **X** on the number in the emoji scale that indicates your understanding in relation to today's learning target.

I can decompose composite figures into triangles and quadrilaterals to find the area.



2. Provide evidence to explain your self-assessment.

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